

Last time: $f: X \rightarrow Y$ continuous $\Leftrightarrow \forall U \subseteq Y$ open,
 $f^{-1}(U) \subseteq X$ open

Cor The composition of continuous functions is continuous:

If X, Y, Z are metric spaces and $f: X \rightarrow Y$,
 $g: Y \rightarrow Z$ are continuous, then $g \circ f: X \rightarrow Z$ is continuous.

Pf: Fix $U \subseteq Z$ open. Want: $(g \circ f)^{-1}(U)$ open in X .

As f, g are continuous,

$$U \text{ open} \Rightarrow g^{-1}(U) \text{ open} \Rightarrow f^{-1}(g^{-1}(U)) \text{ open}$$

Note that $(g \circ f)^{-1}(U) = f^{-1}(g^{-1}(U))$:

$$\begin{aligned} x \in (g \circ f)^{-1}(U) &\Leftrightarrow g(f(x)) \in U \\ &\Leftrightarrow f(x) \in g^{-1}(U) \\ &\Leftrightarrow x \in f^{-1}(g^{-1}(U)). \quad \square \end{aligned}$$

Thm Let X, Y be metric spaces and $f: X \rightarrow Y$.

If f is continuous and $C \subseteq X$ is connected, then $f(C)$ is connected in Y .

Pf: By Lecture 28, it suffices to prove that $f(C)$ is connected in $f(C)$. Consider the function $f: C \rightarrow f(C)$. This is still continuous: $\forall \varepsilon > 0, \exists \delta > 0$ s.t.

$$x \in C, \quad 0 < d_X(x, x_0) < \delta \Rightarrow d_Y(f(x), f(x_0)) < \varepsilon$$

The induced metric
on $C \subseteq X$

The induced metric
on $f(C) \subseteq Y$

Suppose $f(C)$ were not connected. Then $\exists A \subseteq f(C)$ s.t. $A \neq \emptyset, f(C)$ and A is both open and closed in $f(C)$ (by Lecture 29). As $f: C \rightarrow f^{-1}(C)$ is continuous, then $f^{-1}(A)$ is both open and closed in C . Moreover, $f^{-1}(A) \neq \emptyset$ and $f^{-1}(A) \neq C$ since $A \neq \emptyset$ and $A \neq f(C)$. This contradicts that C is connected. $\square$

• Taking $X = \mathbb{R} = Y$, we get:

Cor (Intermediate value theorem) If $f: [a, b] \rightarrow \mathbb{R}$ is continuous and $c \in \mathbb{R}$ satisfies $f(a) < c < f(b)$ or $f(a) > c > f(b)$, then $\exists x_0 \in (a, b)$ s.t. $f(x_0) = c$.

Pf: $[a, b]$ interval $\xRightarrow{\text{lec 29}}$ $[a, b]$ connected
 $\xRightarrow{\text{Thm}}$ $f([a, b])$ connected
 $\xRightarrow{\text{lec 30}}$ $f([a, b])$ is an interval.

As c lies between $f(a)$ and $f(b)$ and $f(a), f(b)$ are in the interval $f([a, b])$, then $c \in f([a, b])$ as well. So $\exists x_0 \in [a, b]$ s.t. $f(x_0) = c$. Note that $x_0 \neq a, b$ since $c \neq f(a), f(b)$. $\square$

• Also applies to $Y = \mathbb{R}^n$:

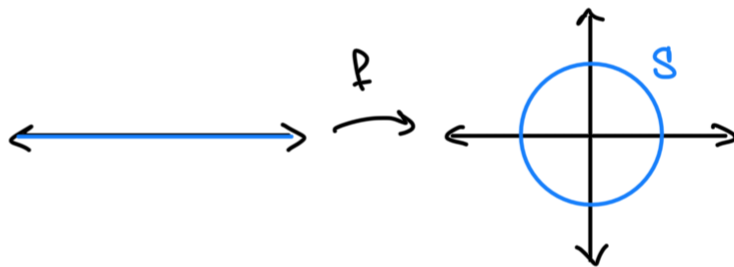
Ex The unit circle $S = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$ is connected, because:

• $f: \mathbb{R} \rightarrow \mathbb{R}^2$ is continuous

$$\theta \mapsto (\cos \theta, \sin \theta)$$

• $\mathbb{R}$ is connected

• $f(\mathbb{R}) = S$



• This allows us to prove that many new sets are connected:

Prop Let (X, d) be a metric space. If X is path connected, i.e.

$$\forall a, b \in X, \quad \exists f: [0, 1] \rightarrow X \text{ continuous s.t.} \\ f(0) = a \text{ and } f(1) = b,$$

then X is connected.

Pf: By HW11 #3, it suffices to show:

$$\forall a, b \in X, \quad \exists C \text{ connected s.t. } a, b \in C.$$

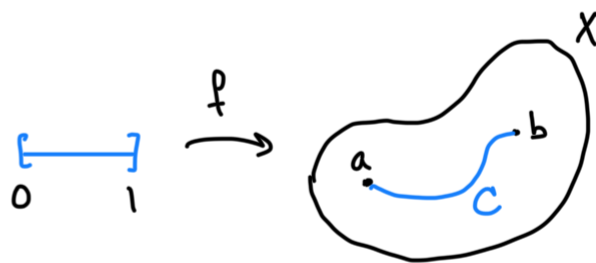
Fix $a, b \in X$. We know $\exists f: [0, 1] \rightarrow X$ continuous s.t. $f(0) = a$ and $f(1) = b$.

Then $C = f([0, 1])$ satisfies:

• $a = f(0) \in C$

• $b = f(1) \in C$

• C is connected: since f is continuous and $[0, 1]$ is connected.



□

Thm Let X, Y be metric spaces and $f: X \rightarrow Y$.

If f is continuous and K is compact, then $f(K)$ is compact.

Pf: Let $\{U_i\}_{i \in I}$ be an open cover of $f(K)$ in Y .

Then $\{f^{-1}(U_i)\}_{i \in I}$ satisfies:

- $f^{-1}(U_i)$ open $\forall i \in I$:
since $U_i \subseteq Y$ open and f is continuous

- $K \subseteq \bigcup_{i \in I} f^{-1}(U_i)$: since

$$x \in K \Rightarrow f(x) \in f(K) \subseteq \bigcup_{i \in I} U_i$$

$$\Rightarrow \exists i_0 \in I \text{ s.t. } f(x) \in U_{i_0}$$

$$\Rightarrow x \in f^{-1}(U_{i_0}) \subseteq \bigcup_{i \in I} f^{-1}(U_i)$$

So $\{f^{-1}(U_i)\}_{i \in I}$ is an open cover of K . As K is compact, $\exists$ a finite subcover $\{f^{-1}(U_i)\}_{i \in J}$.

$$\text{Claim: } f(K) \subseteq \bigcup_{i \in J} U_i.$$

$$y \in f(K) \Rightarrow \exists x \in K \text{ s.t. } f(x) = y$$

$$\Rightarrow x \in K \subseteq \bigcup_{i \in J} f^{-1}(U_i)$$

$$\Rightarrow \exists i_0 \in J \text{ s.t. } x \in f^{-1}(U_{i_0})$$

$$\Rightarrow y = f(x) \in U_{i_0}.$$

As $\{U_i\}_{i \in I}$ was an arbitrary open cover of K , we conclude $f(K)$ is compact. $\square$

